

KYRA, Russian Federation

WMO: 309490

Lat: 49.567N

Lon: 111.967E

Elev: 2979

StdP: 13.18

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.5	-24.6	-33.5	0.9	-26.3	-29.5	1.2	-22.9	14.3	4.7	12.3	5.5	0.2	0	0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	85.4	61.5	81.5	60.6	78.0	59.4	65.7	77.5	64.0	75.0	62.5	72.6	4.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.2	93.5	69.5	60.4	87.5	67.9	58.6	82.0	66.6	32.4	77.1	31.0	75.2	29.7	72.6	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
15.0	12.2	10.0	DB	-32.4	91.6	4.9	5.0	-35.9	95.3	-38.8	98.2	-41.5	101.0	-45.1	104.7	
			WB	-32.7	68.8	4.9	1.8	-36.2	70.1	-39.0	71.1	-41.8	72.1	-45.3	73.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.0	-5.3	3.2	19.5	36.4	49.4	60.9	65.3	61.3	49.4	32.8	11.6	-2.1
	DBStd	26.19	8.57	9.70	10.33	8.10	7.62	6.16	4.70	5.60	7.13	8.64	11.31	10.11
	HDD50	7893	1714	1312	946	412	106	4	0	2	96	534	1152	1615
	HDD65	12154	2179	1732	1411	857	485	150	52	139	469	998	1602	2080
	CDD50	1327	0	0	0	5	87	330	474	353	77	1	0	0
	CDD65	115	0	0	0	0	1	26	61	26	1	0	0	0
	CDH74	1831	0	0	0	2	126	564	694	396	49	0	0	0
	CDH80	516	0	0	0	0	33	178	197	100	8	0	0	0
Wind	WSAvg	3.0	2.1	2.6	3.3	4.6	4.9	3.2	2.5	2.5	3.0	2.9	2.4	2.2
Precipitation	PrecAvg	14.8	0.1	0.1	0.1	0.5	1.0	2.3	4.8	3.9	1.6	0.4	0.1	0.1
	PrecMax	22.7	0.6	0.3	0.4	1.4	3.6	5.4	10.6	7.6	4.7	1.5	0.6	0.3
	PrecMin	8.1	0.0	0.0	0.0	0.0	0.1	0.3	0.9	0.9	0.1	0.0	0.0	0.0
	PrecStd	3.4	0.1	0.1	0.1	0.4	0.8	1.3	2.1	1.8	1.1	0.4	0.2	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.7	38.3	54.6	70.0	85.0	92.0	91.5	87.2	79.9	65.1	44.3	28.0
		MCWB	18.5	29.2	38.9	46.8	56.3	62.3	64.8	62.6	56.1	46.4	34.5	23.2
	2%	DB	17.5	30.3	48.0	63.7	77.4	84.7	84.9	82.6	73.1	58.2	37.6	21.8
		MCWB	14.0	23.9	36.0	45.8	53.5	60.0	63.1	61.7	54.6	43.5	30.9	18.2
	5%	DB	13.1	25.6	42.0	58.0	72.3	80.0	81.0	78.1	69.0	52.8	33.0	17.4
		MCWB	10.4	20.5	32.0	42.5	51.5	58.5	62.5	60.4	53.0	40.6	27.5	14.6
	10%	DB	8.9	20.6	37.0	53.0	67.5	76.2	77.3	74.2	64.3	47.6	28.8	13.4
		MCWB	6.9	16.6	29.0	39.8	49.0	57.3	61.6	59.6	51.3	37.8	24.4	11.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	29.3	40.0	49.7	58.1	65.4	68.4	67.8	60.1	47.8	34.6	22.9
		MCDB	22.8	37.6	53.3	67.0	78.1	82.5	81.8	79.3	71.0	63.3	42.9	27.5
	2%	WB	14.1	24.3	35.9	46.2	55.1	62.7	66.4	65.0	57.0	44.1	31.1	18.2
		MCDB	17.4	30.3	47.6	61.8	75.0	76.9	78.6	75.1	68.1	57.2	37.5	21.6
	5%	WB	10.4	20.6	32.3	43.2	52.3	60.9	64.8	63.2	54.9	41.0	27.7	14.6
		MCDB	13.0	25.7	41.5	56.8	69.0	74.7	76.1	73.0	65.6	52.2	32.7	17.3
	10%	WB	6.9	16.8	29.3	40.2	50.2	59.2	63.4	61.3	52.4	38.2	24.6	11.1
		MCDB	8.9	20.6	36.6	52.0	65.3	72.1	73.5	70.6	63.1	46.9	28.7	13.3
Mean Daily Temperature Range	5% DB	MDBR	22.2	28.3	26.6	25.4	26.7	25.0	21.3	22.9	25.0	23.6	21.8	20.2
		MCDBR	25.0	34.4	34.0	35.9	38.1	35.0	29.7	31.2	34.4	33.6	25.2	22.3
		MCWBR	22.1	28.2	22.9	20.9	18.6	14.1	12.0	12.9	16.8	19.7	19.1	19.4
	5% WB	MCDBR	24.8	34.2	32.7	33.5	33.8	27.1	23.3	23.7	28.7	31.3	24.7	22.4
		MCWBR	22.0	28.2	22.3	20.4	17.6	12.4	11.0	11.3	14.6	18.8	18.9	19.5
Clear-Sky Solar Irradiance	taub	0.250	0.275	0.314	0.377	0.443	0.440	0.406	0.377	0.332	0.296	0.268	0.252	
	taud	2.316	2.288	2.273	2.179	2.020	2.144	2.307	2.366	2.438	2.459	2.380	2.282	
	Ebn at Noon	259	278	284	274	259	261	269	272	275	268	249	238	
	Edh at Noon	21	28	34	43	52	46	39	35	29	24	20	19	
All-Sky Solar Radiation	RadAvg	544	919	1406	1720	1917	1923	1799	1617	1317	896	559	420	
	RadStd	19	30	55	109	115	120	127	117	88	42	24	28	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-3.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	-2.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page